Brief Note — Power BI Test: F&B Distribution Analysis

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1. Data Cleaning Steps

Schema Standardization: Loaded 2 CSV transaction files with different schemas — standardized column names ("Order Date" → "OrderDate"), date formats (DD/MM/YYYY → Date type via UK locale), and discount formats (string "15%" → decimal 0.15). Appended into a single fact table (1,352 rows). February's extra PromotionCode column preserved (null for January).

Data Quality Cleaning: ProductID: Trimmed whitespace + UPPERCASE (~20 rows fixed). Region: Mapped 12 inconsistent variants → 3 values (North, Central, South). Quantity: Replaced 12 null values with 0. Duplicates: Removed 7 exact duplicate rows (ETL errors). Final: 1,345 rows.

Product Merge: LEFT OUTER JOIN with Product_Master to retrieve Category, Brand, UnitCost. Chose Left Outer (not Inner) to preserve all transactions including orphans.

Orphan Orders: 17 transactions referencing non-existent ProductIDs (SKU-2xxx series, ~1.3%). Kept for audit trail, excluded from financial calculations.

Date Table: Created DAX calculated table (Jan 1 – Feb 28, 2025, 59 rows). Marked as Date Table for time intelligence.

2. Assumptions

- Negative quantities (12 transactions, 164 units) = returns — tracked separately via Return Quantity/Return Rate measures
- Null quantities → 0 (do not contribute to revenue). Orphan orders excluded from revenue/COGS but included in order counts
- SalesTarget connected via TREATAS (no physical relationship) — target at Category × Month granularity
- All currency in USD. Only "Active" SKUs considered for reorder alerts

3. Business Questions

Q1: Which category and channel contribute the most revenue? Percentage share?

Spirits leads with $5,946,173 (59.7%), followed by Wine $3,271,568 (32.9%). Together they account for 92.6% of total revenue ($9,957,008). Coffee $517,784 (5.2%), Beer $123,110 (1.2%), Soft Drinks $98,374 (1.0%). By channel, distribution is remarkably even: On-Premise 21.0%, Modern Trade 20.2%, E-commerce 19.8%, General Trade 19.7%, Wholesale 19.3%.

Q2: Top 5 products by gross profit? High revenue but low margin?

Top 5: (1) Crystal Peak Whisky 1000ml $578,696 GP (51.3% margin), (2) Northern Botanic Whisky 750ml $468,306 (50.9%), (3) Highland Reserve Vodka 500ml $447,461 (55.6%), (4) Château Lumière Rose 750ml $426,059 (59.1%), (5) Crystal Peak Rum 500ml $298,861 (43.7%). Notable: Northern Botanic Gin 1000ml has high revenue ($520,471) but low margin (34.9%), and Maison Royale Vodka 750ml ($449,068, 38.8% margin) — both selling below list price.

Q3: Revenue change (%) from January to February? Key drivers?

Revenue grew +11.2% MoM (Jan $4,713,786 → Feb $5,243,222, +$529,436). Drivers: (1) Spirits +13.1% (+$366K, largest absolute contributor), (2) Channel shift — On-Premise surged +38.0%, General Trade +30.2%, while Wholesale dropped -25.5%, (3) February promotions (TET25, NEWYEAR, BULK10) drove 340 promo orders worth $2.3M, (4) Beer +16.9% and Wine +9.4% also contributed.

Q4: Which categories missed their targets? By how much?

All categories exceeded targets significantly. Total actual $9,957,008 vs target $648,400 (1,536% achievement). This suggests targets were set at a much lower scale than actual operations — possibly unit-level targets being compared against revenue. Spirits achieved 2,378% ($5.7M above target), Wine 1,972% ($3.1M above). Even Beer exceeded at 133% (+$30,210). Recommendation: verify target data granularity with operations department.

Q5: Which SKUs need restocking? Consider lead times.

8 Active SKUs are below reorder point — all 8 are URGENT (stock coverage < lead time). Critical (< 1 day): Valle Verde Rose 330ml (6 units, 60d lead time), Dark Harbor Lager 330ml (11 units, 45d), FizzUp Cola 700ml (13 units, 45d), Crystal Peak Whisky 1000ml (44 units, 60d — top profit product!). Severe (~1 day): VoltX Juice 330ml (54 units, 45d), Highlands Origin Roasted Bean 500ml (64 units, 14d), QuickBrew Cold Brew 1000ml (87 units, 60d), Valle Verde White Wine 700ml (104 units, 30d). Highest risk: Crystal Peak Whisky 1000ml — #1 profit product with only 44 units and 60-day lead time from USA. Immediate order required.

Q6: Which customers/regions have concerning overdue payment rates?

Overall overdue rate: 20.6% (75/364 orders), representing $1,273,666 in overdue revenue. Top concerning customers: Saigon Mart (29.2% overdue rate, $131,873), Coastal Bars Ltd (27.6%, $98,650), City Convenience Co. (27.3%, $66,505), Highlands Cafe Chain (27.3%, $193,962). By dollar impact, Highlands Cafe Chain has the largest overdue amount at $193,962 despite moderate rate. Recommendation: implement tiered payment terms — prepayment for customers above 25% overdue rate, shorter net terms for 20-25%.